

How Low-Dimensional Is the S–T–AOU Dependence of Ocean Total Dissolved Inorganic Carbon? A Symbolic Regression Analysis

Anonymous manuscript

Abstract

We investigate how much of the empirical dependence of ocean total dissolved inorganic carbon (TCO_2) on salinity (S), temperature (T), and apparent oxygen utilisation (AOU) can be compressed into a low-dimensional nonlinear representation. We analyse 472,530 quality-controlled GLODAP v2.2023 observations from the Atlantic, Indian, Pacific, and Southern Ocean basins. Symbolic regression comprises 48 searches across four operator spaces and three random seeds, yielding 599 candidate expressions. We compare these candidates with predefined linear, rank-1 quadratic, rank-2 quadratic, full quadratic, and cubic model families.

A rank-1 quadratic model with five fitted coefficients, $\text{TCO}_2 = (\alpha S + \beta T + \gamma \text{AOU}/100 + \delta)^2 + \epsilon$, captures substantial predictive structure. In temporal validation its RMSE is 20.17, 14.48, 16.00, and 10.58 $\mu\text{mol kg}^{-1}$ in the Atlantic, Indian, Pacific, and Southern Ocean, respectively. Rank-2 models do not provide a meaningful improvement: they are worse in the Atlantic and Indian and improve RMSE by only 0.29 and 0.04 $\mu\text{mol kg}^{-1}$ in the Pacific and Southern Ocean. More flexible symbolic candidates reduce validation RMSE by approximately 3–12% relative to rank-1, but require two- to three-fold greater PySR expression complexity. Rank-1 candidates recur in 24 of 48 basin–operator–space–seed combinations; recurrence is strongest in some operator spaces and is especially dependent on explicit squaring in the Atlantic.

Temporal, spatial-blocked, cruise-blocked, and post-2018 external evaluations show that generalization depends on basin and validation regime. The rank-1 structure is therefore best interpreted as an efficient low-complexity empirical projection, not as a universally optimal equation or a physical law. AAIW and Southern Ocean abyssal regimes remain important failure domains, with Southern Ocean abyssal global-fit $R^2 = -0.174$ and locally fitted $R^2 = 0.786$. These results support substantial but non-universal compressibility of the empirical S – T –AOU relationship.

Keywords: total dissolved inorganic carbon; symbolic regression; rank-1 quadratic; GLODAP; model complexity; ocean carbon

1 Introduction

Total dissolved inorganic carbon (TCO_2) is central to ocean carbon-cycle assessments and air–sea CO_2 exchange studies (Sarmiento and Gruber, 2006). Empirical prediction from hydrographic and biogeochemical variables is useful for gap filling, quality control, and regional carbon analyses (Takahashi et al., 1993; Key et al., 2004). Linear and polynomial regressions are interpretable but impose restrictive functional forms, whereas flexible machine-learning methods can be difficult to interpret (Goyet and Davis, 1991; Millero, 1998; Broullón et al., 2019).

Symbolic regression offers a structural search over mathematical expressions. The purpose of the present study is not to declare a new equation or a new physical law. Instead, we ask:

How low-dimensional is the empirical S – T –AOU dependence of ocean TCO_2 , and how much predictive accuracy is retained as structural constraints become stronger?

The central hypothesis is that a substantial fraction of the predictive structure can be represented by a low-dimensional nonlinear projection. We test this hypothesis with predefined model families, symbolic-regression Pareto libraries, multiple operator spaces,

Table 1: Sample sizes by basin and temporal split.

Basin	N_total	N_train	N_validation	N_external
Atlantic	147448	125762	11436	10250
Indian	40391	32498	3764	4129
Pacific	187443	138202	31791	17450
Southern Ocean	97248	82045	6483	8720
TOTAL	472530	378507	53474	40549

repeated seeds, temporal validation, spatial and cruise blocking, and a locked post-2018 external holdout.

2 Data and Methods

2.1 GLODAP data and quality control

We used GLODAP v2.2023 (Lauvset et al., 2023), extracting salinity, temperature, AOU, TCO_2 , latitude, longitude, depth, year, region, cruise, and station. Observations were retained when all required variables were finite and $25 < S < 42$, $-2.5 < T < 35^\circ\text{C}$, $1700 < \text{TCO}_2 < 2600 \mu\text{mol kg}^{-1}$, and $\text{AOU} > -50 \mu\text{mol kg}^{-1}$.

2.2 Basins and temporal split

The Atlantic was defined as region 1 north of 35°S , the Indian as region 16 north of 35°S , the Pacific as region 8 north of 35°S , and the Southern Ocean as latitude south of 35°S . The definitions are mutually exclusive. Training used observations before 2015, temporal validation/model selection used 2015–2017, and the locked external holdout used observations from 2018 onward. No external metric was used for candidate selection, operator selection, or manuscript conclusions.

2.3 Model families

The model hierarchy was:

Table 2: Predictor and target distributions by basin.

Basin	Variable	N	Mean	SD	Q25	Median	Q75
Unclassified	salinity	512384	34.665	0.899	34.483	34.694	34.926
Unclassified	temperature	512384	6.707	7.430	1.626	3.562	10.121
Unclassified	aou	512384	101.873	85.448	29.322	85.784	163.690
Unclassified	tco2	512384	2187.727	104.867	2127.000	2182.000	2261.300
Unclassified	depth	512384	1213.613	1380.175	139.000	613.000	1943.925

1. mean baseline;
2. linear regression, with four fitted parameters;
3. rank-1 quadratic,

$$\text{TCO}_2 = (\alpha S + \beta T + \gamma \text{AOU}/100 + \delta)^2 + \epsilon, \quad (1)$$

- with five fitted parameters and effective quadratic rank one;
4. rank-2 quadratic, $\text{TCO}_2 = z_1^2 + z_2^2 + \epsilon$, with nine fitted parameters;
5. full second-order polynomial, with ten fitted parameters;
6. cubic polynomial, with twenty fitted parameters; and
7. frozen symbolic candidates selected using temporal validation only.

PySR expression complexity is an expression-tree measure and is distinct from fitted-parameter count. For example, the rank-1 quadratic has five fitted coefficients and PySR expression complexity 8.

2.4 Symbolic regression and recurrence

We ran four operator spaces: Search A (+, −, ×, ÷); Search B adding exp and log; Search C adding square and root operators; and Search D with a broader scientifically reasonable set. Each basin–operator-space configuration was run with seeds 42, 123, and 456. All 48 searches completed successfully. The complete candidate library was retained. Candidate selection used temporal validation RMSE only. Rank-1 family classification is syntactic and therefore treated as a heuristic structural classification rather than an algebraic proof.

Table 3: Symbolic-regression search configurations.

Search	Binary operators	Unary operators	Max size	Populations	Population size	Iterations
SearchA_basic	+, -, *, /		30	30	50	100
SearchB_explog	+, -, *, /	exp, log	30	30	50	100
SearchC_sqrt_square	+, -, *, /	sqrt, square, exp, log	30	30	50	100
SearchD_broad	+, -, *, /	sqrt, square, exp, log, abs, cube, cbrt	35	40	60	150

2.5 Blocked validation

To assess spatial generalization, we used GroupKFold over 5° latitude by 10° longitude blocks. To assess sampling-event generalization, we used cruise identifiers when available. Both analyses excluded all observations from 2018 onward. Train and validation groups were checked for disjointness, and fold-level predictions were retained. The frozen symbolic candidate was not refit or selected inside blocked folds.

2.6 Dependence-aware inference

Paired absolute-error differences were assessed with spatial-block bootstrap resampling. Equivalence margins were $\delta = 1, 2, 5$, and $10 \mu\text{mol kg}^{-1}$. We distinguish statistical detectability from practical magnitude and do not interpret naive observation-level TOST as evidence of dependence-aware equivalence.

2.7 Depth, water mass, and derivatives

Depth bins were 0–100, 100–500, 500–1000, 1000–3000, and 3000–7000 m. Threshold-based classifications were used for Surface, Thermocline, AAIW, NADW, and AABW. For $z = \alpha S + \beta T + \gamma \text{AOU}/100 + \delta$, derivatives are

$$\frac{\partial \text{TCO}_2}{\partial S} = 2\alpha z, \quad (2)$$

$$\frac{\partial \text{TCO}_2}{\partial T} = 2\beta z, \quad (3)$$

$$\frac{\partial \text{TCO}_2}{\partial \text{AOU}} = \frac{2\gamma}{100} z. \quad (4)$$

These derivatives describe the fitted empirical surface; their signs do not establish causality.

Table 4: Final model hierarchy across temporal validation and external holdout.

basin	model	parameter_count	expression_complexity	validation_RMSE	validation_MAE	validation_R2	external_RMSE	external_MAE	external_R2	validation_delta_vs_rank1
Atlantic	mean	1.000	1	66.879	49.937	-0.017	62.940	48.806	-0.001	46.708
Atlantic	linear	4.000	4	21.591	14.666	0.894	18.538	13.134	0.913	1.419
Atlantic	rank1	5.000	8	20.172	14.926	0.907	17.672	13.170	0.921	0.000
Atlantic	rank2	9.000	16	22.155	14.990	0.888	19.586	12.579	0.903	1.984
Atlantic	full_quadratic	10.000	20	23.406	15.570	0.875	19.621	12.528	0.903	3.235
Atlantic	cubic	20.000	30	22.598	14.802	0.884	18.010	12.091	0.918	2.427
Atlantic	frozen_symbolic	NaN	15	18.998	13.292	0.918	13.125	9.502	0.956	-1.174
Indian	mean	1.000	1	122.473	105.708	-0.052	108.518	95.689	-0.102	107.991
Indian	linear	4.000	4	14.960	11.748	0.984	19.137	14.208	0.966	0.478
Indian	rank1	5.000	8	14.483	11.056	0.985	18.413	13.442	0.968	0.000
Indian	rank2	9.000	16	14.737	11.139	0.985	18.858	13.813	0.967	0.254
Indian	full_quadratic	10.000	20	13.729	10.680	0.987	14.922	11.457	0.979	-0.753
Indian	cubic	20.000	30	11.827	9.126	0.990	16.526	11.688	0.974	-2.655
Indian	frozen_symbolic	NaN	25	12.800	9.702	0.989	16.797	11.744	0.974	-1.682
Pacific	mean	1.000	1	131.396	116.426	-0.001	140.892	129.001	-0.046	115.394
Pacific	linear	4.000	4	17.769	13.414	0.982	17.196	13.403	0.984	1.768
Pacific	rank1	5.000	8	16.002	11.089	0.985	13.821	9.636	0.990	0.000
Pacific	rank2	9.000	16	15.712	11.096	0.986	14.087	10.051	0.990	-0.289
Pacific	full_quadratic	10.000	20	15.504	10.704	0.986	14.210	9.554	0.989	-0.497
Pacific	cubic	20.000	30	14.244	9.341	0.988	12.451	8.448	0.992	-1.758
Pacific	frozen_symbolic	NaN	20	15.488	10.706	0.986	13.531	8.995	0.990	-0.514
Southern Ocean	mean	1.000	1	59.190	50.841	-0.000	51.755	47.214	-0.156	48.614
Southern Ocean	linear	4.000	4	10.664	7.478	0.968	8.930	6.076	0.966	0.089
Southern Ocean	rank1	5.000	8	10.575	7.549	0.968	8.985	5.985	0.965	0.000
Southern Ocean	rank2	9.000	16	10.540	7.607	0.968	8.895	5.985	0.966	-0.035
Southern Ocean	full_quadratic	10.000	20	10.461	7.603	0.969	8.808	5.887	0.967	-0.115
Southern Ocean	cubic	20.000	30	10.942	7.418	0.971	8.687	5.884	0.967	-0.533
Southern Ocean	frozen_symbolic	NaN	21	9.988	7.032	0.972	8.597	5.664	0.968	-0.588

3 Results

3.1 Data and baseline reproduction

Quality control retained 472,530 observations assigned to the four basins. The baseline reproduction matched the previous analysis within rounding: temporal-validation RMSE was 20.17, 14.48, 16.00, and 10.58 $\mu\text{mol kg}^{-1}$ for the rank-1 model in the Atlantic, Indian, Pacific, and Southern Ocean, respectively. The full quadratic RMSE values were 23.41, 13.73, 15.50, and 10.46 $\mu\text{mol kg}^{-1}$.

3.2 Model hierarchy and rank comparison

Rank-1 improved on linear regression in all basins, but it was not uniformly the most accurate parametric model. It outperformed full quadratic in the Atlantic, while full quadratic was better by 0.75, 0.50, and 0.11 $\mu\text{mol kg}^{-1}$ in the Indian, Pacific, and Southern Ocean, respectively. Cubic models achieved lower temporal-validation RMSE in the Indian, Pacific, and Southern Ocean, but at four times the fitted-parameter count of rank-1.

3.3 Rank-1 recurrence and operator-space dependence

Rank-1 candidates comprised 86 of 599 retained candidates and occurred in 24 of 48 basin-operator-space-seed cells. Thus recurrence was 50% at the cell level, although all

Table 5: Model family comparison. Validation and external holdout metrics.

Basin	Model_name	Parameters	RMSE_val	MAE_val	R2_val	RMSE_ext	R2_ext
Atlantic	Baseline Mean	1	66.879	49.937	-0.017	62.940	-0.001
Atlantic	Linear Regression	4	21.591	14.666	0.894	18.538	0.913
Atlantic	Rank-1 Quadratic	5	20.172	14.926	0.907	17.672	0.921
Atlantic	Rank-2 Quadratic	9	22.155	14.990	0.888	19.586	0.903
Atlantic	Full Second-Order Polynomial	10	23.406	15.570	0.875	19.621	0.903
Atlantic	Cubic Polynomial	20	22.598	14.802	0.884	18.010	0.918
Indian	Baseline Mean	1	122.473	105.708	-0.052	108.518	-0.102
Indian	Linear Regression	4	14.960	11.748	0.984	19.137	0.966
Indian	Rank-1 Quadratic	5	14.483	11.056	0.985	18.413	0.968
Indian	Rank-2 Quadratic	9	14.737	11.139	0.985	18.858	0.967
Indian	Full Second-Order Polynomial	10	13.729	10.680	0.987	14.922	0.979
Indian	Cubic Polynomial	20	11.827	9.126	0.990	16.526	0.974
Pacific	Baseline Mean	1	131.396	116.426	-0.001	140.892	-0.046
Pacific	Linear Regression	4	17.769	13.414	0.982	17.196	0.984
Pacific	Rank-1 Quadratic	5	16.002	11.089	0.985	13.821	0.990
Pacific	Rank-2 Quadratic	9	15.712	11.096	0.986	14.087	0.990
Pacific	Full Second-Order Polynomial	10	15.504	10.704	0.986	14.210	0.989
Pacific	Cubic Polynomial	20	14.244	9.341	0.988	12.451	0.992
Southern Ocean	Baseline Mean	1	59.190	50.841	-0.001	51.755	-0.156
Southern Ocean	Linear Regression	4	10.664	7.478	0.968	8.930	0.966
Southern Ocean	Rank-1 Quadratic	5	10.575	7.549	0.968	8.985	0.965
Southern Ocean	Rank-2 Quadratic	9	10.540	7.607	0.968	8.895	0.966
Southern Ocean	Full Second-Order Polynomial	10	10.461	7.603	0.969	8.808	0.967
Southern Ocean	Cubic Polynomial	20	10.042	7.418	0.971	8.687	0.967

118 four basins contained at least one rank-1 candidate and every operator space produced
 119 rank-1 candidates in aggregate. In the Atlantic, rank-1 occurred only in the search space
 120 with explicit squaring. This result supports recurrence across the broader experiment but
 121 rejects operator-space invariance.

Table 6: Rank-1 recurrence by basin, operator space, and seed. R1 C denotes PySR expression complexity.

Basin	Search	Seed	R1	Pareto	N	R1 RMSE	R1 C
Atlantic	SearchA_basic	42	False	False	0	NaN	NaN
Atlantic	SearchA_basic	123	False	False	0	NaN	NaN
Atlantic	SearchA_basic	456	False	False	0	NaN	NaN
Atlantic	SearchB_explog	42	False	False	0	NaN	NaN
Atlantic	SearchB_explog	123	False	False	0	NaN	NaN
Atlantic	SearchB_explog	456	False	False	0	NaN	NaN
Atlantic	SearchC_sqrt_square	42	True	False	3	21.442	16.000
Atlantic	SearchC_sqrt_square	123	True	False	1	41.185	4.000

Continued on next page

Table 6: Rank-1 recurrence by basin, operator space, and seed. R1 C denotes PySR expression complexity.

Basin	Search	Seed	R1	Pareto	N	R1 RMSE	R1 C
Atlantic	SearchC_sqrt_square	456	True	False	1	35.555	12.000
Atlantic	SearchD_broad	42	False	False	0	NaN	NaN
Atlantic	SearchD_broad	123	False	False	0	NaN	NaN
Atlantic	SearchD_broad	456	False	False	0	NaN	NaN
Indian	SearchA_basic	42	False	False	0	NaN	NaN
Indian	SearchA_basic	123	False	False	0	NaN	NaN
Indian	SearchA_basic	456	True	False	1	14.175	25.000
Indian	SearchB_explog	42	False	False	0	NaN	NaN
Indian	SearchB_explog	123	False	False	0	NaN	NaN
Indian	SearchB_explog	456	True	False	1	12.800	25.000
Indian	SearchC_sqrt_square	42	True	False	1	65.770	4.000
Indian	SearchC_sqrt_square	123	True	False	7	14.150	21.000
Indian	SearchC_sqrt_square	456	True	False	2	30.544	8.000
Indian	SearchD_broad	42	True	False	5	18.340	16.000
Indian	SearchD_broad	123	True	False	2	26.774	8.000
Indian	SearchD_broad	456	False	False	0	NaN	NaN
Pacific	SearchA_basic	42	False	False	0	NaN	NaN
Pacific	SearchA_basic	123	False	False	0	NaN	NaN
Pacific	SearchA_basic	456	False	False	0	NaN	NaN
Pacific	SearchB_explog	42	False	False	0	NaN	NaN
Pacific	SearchB_explog	123	False	False	0	NaN	NaN
Pacific	SearchB_explog	456	False	False	0	NaN	NaN
Pacific	SearchC_sqrt_square	42	True	False	12	15.601	22.000
Pacific	SearchC_sqrt_square	123	True	False	3	31.097	10.000
Pacific	SearchC_sqrt_square	456	True	False	7	16.028	16.000
Pacific	SearchD_broad	42	True	False	4	17.649	12.000

Continued on next page

Table 6: Rank-1 recurrence by basin, operator space, and seed. R1 C denotes PySR expression complexity.

Basin	Search	Seed	R1	Pareto	N	R1 RMSE	R1 C
Pacific	SearchD_broad	123	True	False	11	15.550	22.000
Pacific	SearchD_broad	456	False	False	0	NaN	NaN
Southern Ocean	SearchA_basic	42	True	False	2	9.988	21.000
Southern Ocean	SearchA_basic	123	False	False	0	NaN	NaN
Southern Ocean	SearchA_basic	456	True	False	1	10.827	19.000
Southern Ocean	SearchB_explog	42	True	False	1	10.622	23.000
Southern Ocean	SearchB_explog	123	False	False	0	NaN	NaN
Southern Ocean	SearchB_explog	456	False	False	0	NaN	NaN
Southern Ocean	SearchC_sqrt_square	42	True	False	2	11.863	20.000
Southern Ocean	SearchC_sqrt_square	123	True	False	3	17.379	8.000
Southern Ocean	SearchC_sqrt_square	456	True	False	3	20.527	8.000
Southern Ocean	SearchD_broad	42	True	False	1	42.012	4.000
Southern Ocean	SearchD_broad	123	True	False	10	11.612	14.000
Southern Ocean	SearchD_broad	456	True	False	2	12.278	17.000

3.4 Accuracy–complexity tradeoff

The best symbolic candidates improved temporal-validation RMSE relative to rank-1 by approximately 5.8% in the Atlantic, 11.6% in the Indian, 3.2% in the Pacific, and 5.6% in the Southern Ocean. Their expression complexities were 15, 25, 20, and 21, respectively, compared with rank-1 complexity 8. Rank-1 therefore occupies the low-complexity portion of the frontier but is not the globally most accurate candidate or a demonstrated Pareto knee.

3.5 Temporal, spatial, cruise, and external generalization

Spatial- and cruise-blocked validation used only pre-2018 observations. The results show that rankings can change with the validation regime. For example, in Atlantic spatial

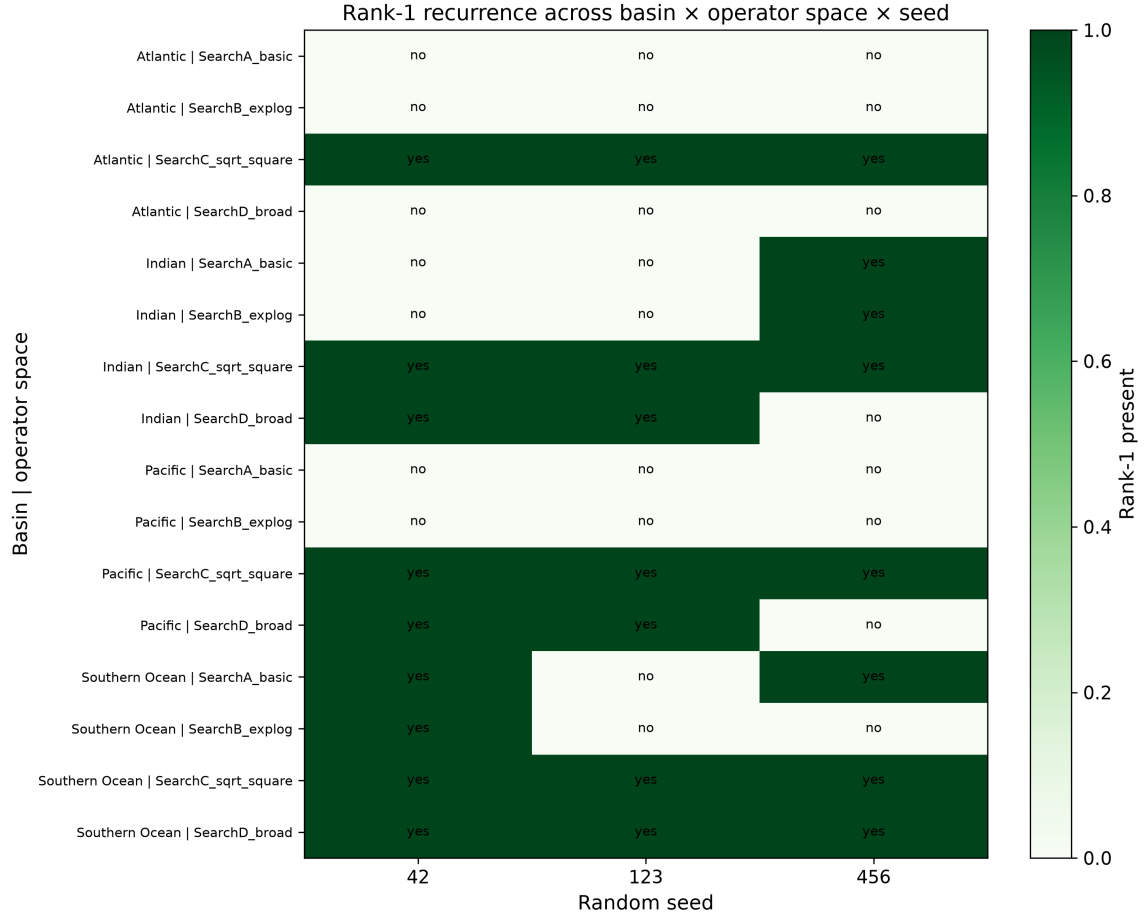


Figure 1: Rank-1 recurrence by basin, operator space, and random seed. Green cells indicate at least one rank-1 candidate in the corresponding search cell.

Table 7: Pareto-frontier summaries and tolerance thresholds.

basin	best_RMSE	best_complexity	rank1_RMSE	rank1_complexity	tolerance	simplest_within_tolerance_RMSE	simplest_within_tolerance_complexity	rank1_within_tolerance
Atlantic	18.998	15	20.172	8	0.010	19.041	13	False
Atlantic	18.998	15	20.172	8	0.020	19.041	13	False
Atlantic	18.998	15	20.172	8	0.050	19.041	13	False
Indian	12.800	25	14.483	8	0.010	12.800	25	False
Indian	12.800	25	14.483	8	0.020	12.963	24	False
Indian	12.800	25	14.483	8	0.050	13.095	20	False
Pacific	15.488	20	16.002	8	0.010	15.488	20	False
Pacific	15.488	20	16.002	8	0.020	15.488	20	False
Pacific	15.488	20	16.002	8	0.050	16.198	15	True
Southern Ocean	9.988	21	10.575	8	0.010	10.080	20	False
Southern Ocean	9.988	21	10.575	8	0.020	10.126	15	False
Southern Ocean	9.988	21	10.575	8	0.050	10.429	12	False

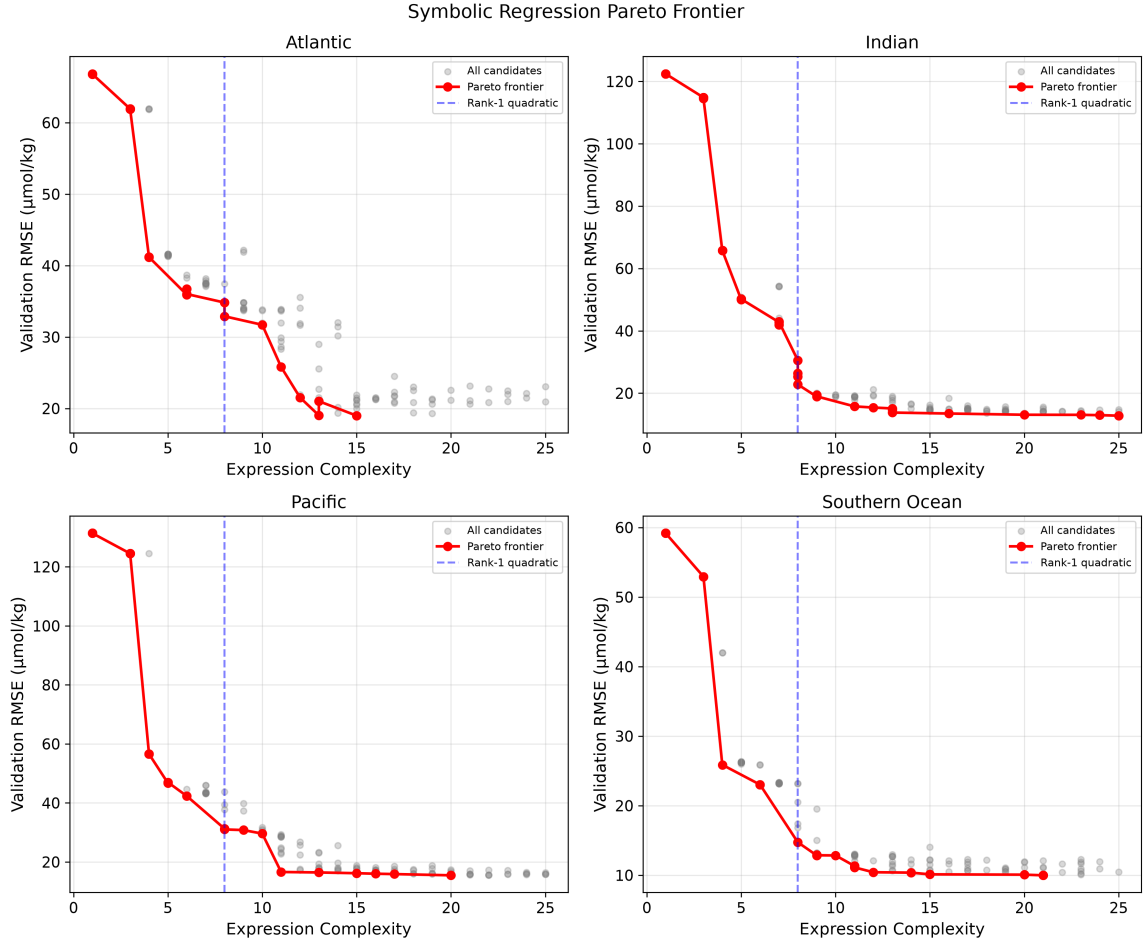


Figure 2: Symbolic-regression validation RMSE versus PySR expression complexity. The rank-1 quadratic is a five-parameter model with expression complexity 8; expression complexity and fitted-parameter count are distinct.

validation, rank-2 RMSE was 15.46, full quadratic 15.55, and rank-1 16.30 $\mu\text{mol kg}^{-1}$; in
cruise validation, rank-2 and full quadratic were again lower than rank-1. In the Indian,
Pacific, and Southern Ocean, the frozen symbolic candidates were competitive or best
under several blocked regimes, but their selection remained based solely on temporal
validation.

Table 8: Spatial- and cruise-blocked validation summary. Full fold diagnostics are in the CSV output.

Basin	Block	Model	Groups	RMSE	RMSE SD	R2
Atlantic	cruise	frozen_symbolic	192	17.519	4.689	0.893
Atlantic	cruise	full_quadratic	192	15.629	2.777	0.914
Atlantic	cruise	linear	192	16.667	4.336	0.903
Atlantic	cruise	rank1	192	16.910	3.886	0.900
Atlantic	cruise	rank2	192	15.579	2.502	0.915
Atlantic	spatial	frozen_symbolic	159	17.927	1.977	0.888
Atlantic	spatial	full_quadratic	159	15.547	2.323	0.915
Atlantic	spatial	linear	159	16.926	2.626	0.900
Atlantic	spatial	rank1	159	16.299	2.110	0.907
Atlantic	spatial	rank2	159	15.464	2.113	0.917
Indian	cruise	frozen_symbolic	40	14.064	3.698	0.984
Indian	cruise	full_quadratic	40	15.375	4.687	0.981
Indian	cruise	linear	40	17.554	4.492	0.975
Indian	cruise	rank1	40	17.394	5.593	0.975
Indian	cruise	rank2	40	16.981	4.927	0.977
Indian	spatial	frozen_symbolic	88	14.354	1.830	0.984
Indian	spatial	full_quadratic	88	14.919	2.553	0.983
Indian	spatial	linear	88	16.989	1.879	0.978
Indian	spatial	rank1	88	16.609	2.093	0.979
Indian	spatial	rank2	88	16.003	1.805	0.980
Pacific	cruise	frozen_symbolic	383	15.520	2.437	0.987

Continued on next page

Table 8: Spatial- and cruise-blocked validation summary. Full fold diagnostics are in the CSV output.

Basin	Block	Model	Groups	RMSE	RMSE SD	R2
Pacific	cruise	full_quadratic	383	16.097	3.730	0.986
Pacific	cruise	linear	383	18.593	1.864	0.982
Pacific	cruise	rank1	383	16.031	2.327	0.986
Pacific	cruise	rank2	383	15.866	2.998	0.986
Pacific	spatial	frozen_symbolic	241	15.634	1.217	0.987
Pacific	spatial	full_quadratic	241	16.428	2.948	0.986
Pacific	spatial	linear	241	18.623	1.048	0.982
Pacific	spatial	rank1	241	16.031	1.280	0.987
Pacific	spatial	rank2	241	16.256	2.547	0.986
Southern Ocean	cruise	frozen_symbolic	114	9.052	0.703	0.978
Southern Ocean	cruise	full_quadratic	114	9.295	0.934	0.976
Southern Ocean	cruise	linear	114	9.418	0.748	0.976
Southern Ocean	cruise	rank1	114	9.225	0.797	0.977
Southern Ocean	cruise	rank2	114	9.195	0.893	0.977
Southern Ocean	spatial	frozen_symbolic	232	9.041	0.863	0.977
Southern Ocean	spatial	full_quadratic	232	9.381	1.617	0.975
Southern Ocean	spatial	linear	232	9.342	0.854	0.976
Southern Ocean	spatial	rank1	232	9.157	0.939	0.977
Southern Ocean	spatial	rank2	232	9.138	1.147	0.976

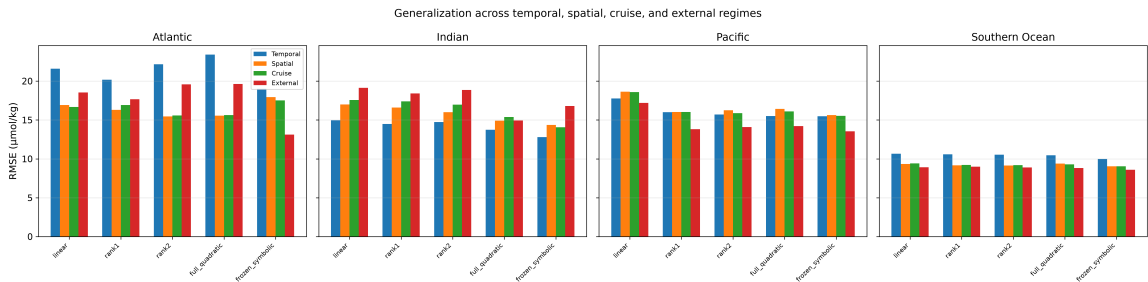


Figure 3: Model RMSE across temporal validation, spatial-blocked validation, cruise-blocked validation, and the locked post-2018 external holdout.

Table 9: Depth-stratified model comparison by basin.

depth_bin	n	RMSE_Rank1	RMSE_FullQuad	R2_Rank1	R2_FullQuad	Improvement_pct	Bias_Rank1	Bias_FullQuad	Basin
0-100m	3469	28.002	36.466	0.758	0.589	23.210	13.199	10.903	Atlantic
100-500m	2477	18.223	16.555	0.781	0.819	-10.075	12.270	10.960	Atlantic
500-1000m	1559	17.617	14.974	0.500	0.639	-17.646	7.516	6.251	Atlantic
1000-3000m	2625	12.303	11.390	0.721	0.760	-8.017	-0.641	0.636	Atlantic
3000-7000m	1306	13.254	14.730	0.784	0.733	10.020	0.275	4.087	Atlantic
0-100m	504	24.713	23.704	0.900	0.908	-4.256	13.500	19.173	Indian
100-500m	961	14.808	14.533	0.953	0.954	-1.891	10.279	10.702	Indian
500-1000m	591	5.699	8.643	0.987	0.971	34.063	-0.416	3.408	Indian
1000-3000m	1181	11.941	10.161	0.701	0.784	-17.516	8.461	7.506	Indian
3000-7000m	527	12.576	10.422	0.466	0.633	-20.669	8.764	5.073	Indian
0-100m	6153	27.212	26.488	0.854	0.862	-2.733	15.345	16.457	Pacific
100-500m	7871	15.821	15.274	0.970	0.972	-3.578	7.207	8.321	Pacific
500-1000m	4619	9.963	9.544	0.982	0.984	-4.395	-4.751	-3.804	Pacific
1000-3000m	8129	8.691	8.761	0.962	0.962	0.805	-0.028	-0.769	Pacific
3000-7000m	5019	10.385	9.405	0.804	0.839	-10.424	3.195	3.853	Pacific
0-100m	1202	17.934	17.438	0.868	0.875	-2.846	7.711	7.309	Southern Ocean
100-500m	1652	10.459	10.638	0.961	0.960	1.688	7.678	8.079	Southern Ocean
500-1000m	975	5.994	7.002	0.980	0.973	14.397	3.136	4.033	Southern Ocean
1000-3000m	1799	6.470	6.102	0.878	0.892	-6.015	3.102	2.806	Southern Ocean
3000-7000m	855	7.447	7.021	-0.174	-0.044	-6.054	5.876	5.479	Southern Ocean

Table 10: Atlantic performance by threshold-based water mass.

n	RMSE	MAE	R2	Bias	NRMSE_pct	Mean_AE	Median_AE	P90_AE	Within_5	Within_10	Within_20	Within_30	water_mass	Model	Basin
4469	26.364	20.294	0.819	13.463	1.266	20.294	17.324	37.192	14.187	28.664	58.223	80.130	Surface	Rank1	Atlantic
1474	14.660	10.982	0.851	6.823	0.674	10.982	8.068	25.818	34.871	58.955	80.258	94.844	Thermocline	Rank1	Atlantic
428	28.426	27.276	-0.115	27.143	1.293	27.276	28.459	36.239	1.636	2.804	17.290	59.346	AAIW	Rank1	Atlantic
2693	9.767	7.574	0.660	-2.701	0.449	7.574	5.983	16.158	42.406	70.702	95.210	99.554	NADW	Rank1	Atlantic
83	29.013	28.261	-15.004	28.261	1.282	28.261	29.333	35.403	0.000	1.205	10.843	54.217	AABW	Rank1	Atlantic
2289	15.607	12.842	0.657	4.222	0.713	12.842	11.435	24.964	21.407	43.032	79.554	95.457	Unclassified	Rank1	Atlantic
4469	33.264	23.317	0.712	11.381	1.597	23.317	18.213	44.578	14.478	28.530	54.755	76.415	Surface	FullQuad	Atlantic
1474	12.197	9.131	0.897	5.541	0.561	9.131	6.573	21.102	39.213	65.468	87.788	98.440	Thermocline	FullQuad	Atlantic
428	22.333	21.342	0.312	21.126	1.016	21.342	21.746	28.508	1.636	4.907	36.449	92.991	AAIW	FullQuad	Atlantic
2693	9.353	7.173	0.688	-0.591	0.430	7.173	5.493	16.103	46.454	73.375	95.284	99.740	NADW	FullQuad	Atlantic
83	36.052	35.427	-23.711	35.427	1.593	35.427	36.381	43.096	0.000	0.000	1.205	14.458	AABW	FullQuad	Atlantic
2289	15.417	12.669	0.666	5.373	0.704	12.669	10.611	25.173	20.358	46.789	79.249	95.893	Unclassified	FullQuad	Atlantic

The locked external holdout produced rank-1 RMSE values of 17.67, 18.41, 13.82, and 8.99 $\mu\text{mol kg}^{-1}$ across the four basins. External rankings were not identical to temporal-validation rankings, reinforcing the distinction between model selection and final evaluation.

3.6 Depth and water-mass dependence

Depth dependence was heterogeneous and sample sizes varied across strata.

In Atlantic AAIW, rank-1 RMSE was approximately 28.4 $\mu\text{mol kg}^{-1}$ with bias +27.1 $\mu\text{mol kg}^{-1}$, compared with full quadratic RMSE approximately 22.3 $\mu\text{mol kg}^{-1}$. This pattern is consistent with omitted state information or coefficient-transfer limitations, but does not establish a causal mechanism.

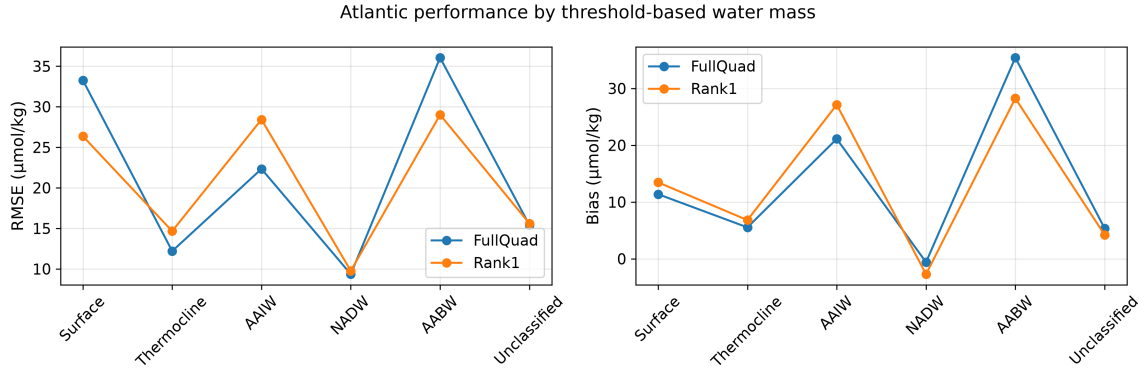


Figure 4: Atlantic RMSE and bias by threshold-based water-mass class. AAIW remains a high-error regime for the rank-1 model.

3.7 Southern Ocean abyssal failure

For the Southern Ocean abyss, the global rank-1 fit gave $R^2 = -0.174$ and RMSE 7.45 $\mu\text{mol kg}^{-1}$, whereas a local abyssal fit gave $R^2 = 0.786$ and RMSE 3.18 $\mu\text{mol kg}^{-1}$. The contrast is consistent with domain shift or coefficient-transfer failure rather than intrinsic unpredictability, but the available analysis does not identify a unique mechanism.

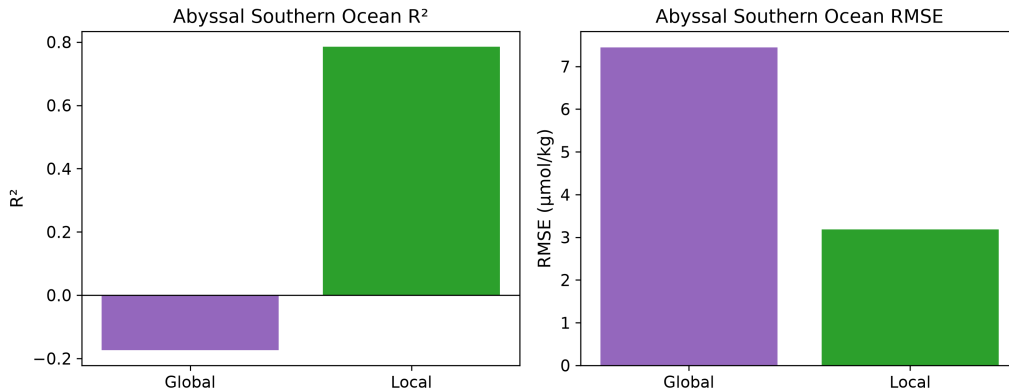


Figure 5: Global versus locally fitted rank-1 performance in the Southern Ocean abyssal regime.

3.8 Derivative structure

Across the observed predictor domains in all four basins, the fitted rank-1 core variable was positive for 100% of observations. Consequently, the fitted derivatives had the expected signs: positive with salinity, negative with temperature, and positive with AOU. These are properties of the fitted empirical surface over the observed domain and should not be

Table 11: Derivative sign analysis across the observed predictor domain.

core_median	core_min	core_max	frac_dS_positive_pct	frac_dT_negative_pct	frac_dA_positive_pct	dS_median	dT_median	dA_median	dS_q25	dS_q75	dT_q25	dT_q75	dA_q25	dA_q75	core_positive_fraction	basin
15.47	-5.38	19.97	99.99	99.99	99.99	45.97	-7.35	0.69	41.59	48.33	-7.73	-6.65	0.62	0.72	99.99	Atlantic
28.72	20.71	30.11	100.00	100.00	100.00	28.78	-9.74	0.74	25.95	29.24	-9.90	-8.79	0.67	0.75	100.00	Indian
22.00	5.56	24.82	100.00	100.00	100.00	51.01	-11.86	0.78	39.24	53.00	-12.46	-9.12	0.60	0.82	100.00	Pacific
20.87	10.70	22.46	100.00	100.00	100.00	30.42	-9.63	0.68	27.66	30.77	-9.74	-8.75	0.62	0.69	100.00	Southern Ocean

Table 12: Dependence-aware model-difference and equivalence results.

n	delta	alpha	mean_AE_diff	SE	tost_p	equiv_by_test	ci_lo	ci_hi	ci_within_bounds	basin	method	n_bootstrap	confidence	mean_diff	boot_std	uses_cluster
11436.000	1.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	2.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	5.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	10.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	-0.395	1.583	NaN	Atlantic	cluster_bootstrap	1000.000	0.950	0.644	0.504	True
3764.000	1.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	2.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	5.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	10.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	-1.050	0.455	NaN	Indian	cluster_bootstrap	1000.000	0.950	-0.376	0.376	True
31791.000	1.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	2.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	5.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	10.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	-0.626	-0.139	NaN	Pacific	cluster_bootstrap	1000.000	0.950	-0.385	0.124	True
6483.000	1.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	2.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	5.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	10.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN	-0.101	0.177	NaN	Southern Ocean	cluster_bootstrap	1000.000	0.950	0.054	0.073	True

interpreted as causal sensitivities outside that domain.

3.9 Dependence-aware equivalence

The spatial-block bootstrap showed basin-specific inference. In particular, the Pacific paired-error interval was entirely negative (approximately $[-0.63, -0.14]$ $\mu\text{mol kg}^{-1}$ for rank-1 minus full quadratic under the stored sign convention), indicating a small but detectable disadvantage for rank-1. We therefore do not claim universal statistical equivalence.

Table 13: Equivalence-margin sensitivity across basins.

n	delta	alpha	mean_AE_diff	SE	tost_p	equiv_by_test	ci_lo	ci_hi	ci_within_bounds	basin	method	n_bootstrap	confidence	mean_diff	boot_std	uses_cluster
11436.000	1.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	2.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	5.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
11436.000	10.000	0.050	0.644	0.101	0.000	True	0.478	0.809	True	Atlantic	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	1.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	2.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	5.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
3764.000	10.000	0.050	-0.376	0.117	0.000	True	-0.569	-0.183	True	Indian	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	1.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	2.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	5.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
31791.000	10.000	0.050	-0.385	0.023	0.000	True	-0.423	-0.347	True	Pacific	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	1.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	2.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	5.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN
6483.000	10.000	0.050	0.054	0.018	0.000	True	0.024	0.084	True	Southern Ocean	naive_tost	NaN	NaN	NaN	NaN	NaN

4 Discussion

4.1 Evidence for low-dimensional empirical structure

The rank-1 model improves over linear regression in every basin and is close to the full quadratic in aggregate temporal performance despite using five fitted coefficients rather than ten. The stronger result is the rank comparison: adding a second quadratic direction produces no meaningful benefit. This supports substantial compressibility of the empirical relationship, not exact one-dimensionality in every validation regime.

4.2 Why rank-1 remains useful despite higher-accuracy models

Higher-complexity symbolic and polynomial models can reduce RMSE, especially in the Indian and Pacific basins. Their gains are purchased with substantially greater expression complexity and fitted-parameter count. Rank-1 is therefore useful as a compact, interpretable approximation, not because it is universally most accurate.

4.3 Operator-space dependence

The 50% cell-level recurrence and Atlantic dependence on explicit squaring show that symbolic discovery is sensitive to the allowed operator space. This is a result about the interaction between data structure and search parameterization; it is not evidence that the rank-1 structure is invariant under arbitrary symbolic representations.

4.4 Domain dependence and generalization

The differences among temporal, spatial, cruise, and external evaluations show that predictive performance depends on sampling structure and domain. AAIW and Southern Ocean abyssal failures are therefore central results. They delimit the range over which a compact basin-level projection can be transferred.

4.5 Physical interpretation

The Hessian establishes rank-1 curvature in predictor space and the derivatives describe the local fitted surface. These facts do not establish a physical mixing axis, causal mechanism, or carbonate-chemistry derivation. The relation is qualitatively compatible with nonlinear carbonate buffering, but that interpretation remains post-hoc and testable rather than demonstrated.

4.6 Limitations

Important limitations include the restricted operator sets, heuristic equation-family classification, dependence among observations, threshold-based water-mass definitions, and the absence of a fully independent global cruise network beyond the available GLODAP cruise identifiers. The post-2018 holdout was locked for evaluation, but the 2015–2017 validation set was used for candidate selection and should not be described as an independent test set.

5 Conclusions

The empirical S–T–AOU dependence of ocean TCO_2 is substantially compressible into a low-dimensional nonlinear representation. A five-parameter rank-1 quadratic captures much of the predictive structure of more flexible models, while increasing quadratic rank from one to two provides little additional predictive benefit. However, rank-1 is not universally optimal, its recurrence depends on operator space, and its performance varies by basin, depth, water mass, and validation regime. The Southern Ocean abyss and AAIW remain important failure domains. The contribution is therefore an evidence-based characterization of empirical low-dimensionality and its limits, not a fundamental ocean-carbon law.

209 Figures

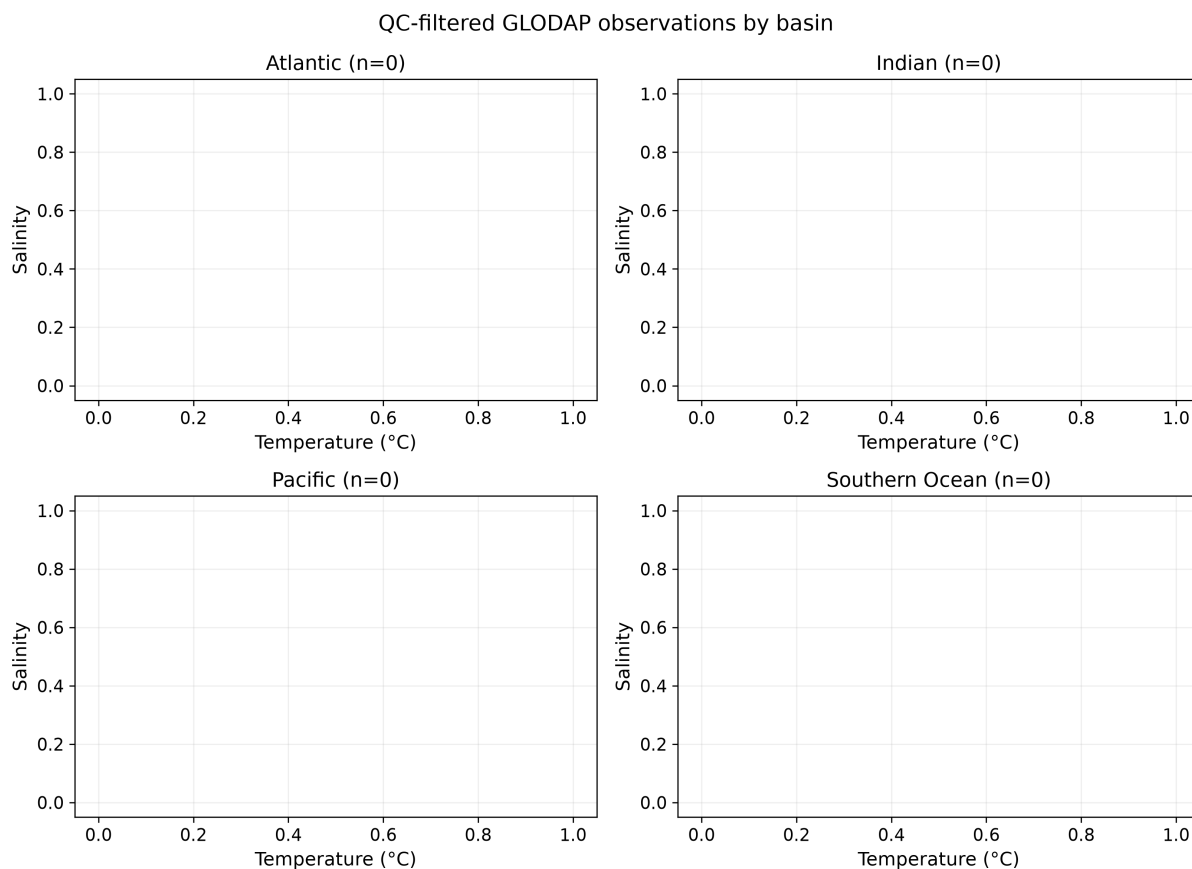


Figure 6: QC-filtered GLODAP observations by basin in temperature–salinity space, colored by TCO_2 .

210 Data and Code Availability

211 GLODAP v2.2023 is available from the Global Ocean Data Analysis Project. The anal-
 212 ysis configuration, environment record, data checksums, generated results, and execution
 213 commands are included in this project. Redistribution of the raw GLODAP file is subject
 214 to the data provider’s terms.

215 References

216 Broullón, D. et al. (2019). A global monthly climatology of total ocean dissolved inorganic
 217 carbon: A neural network approach. *Earth System Science Data*, 11:1429–1447.

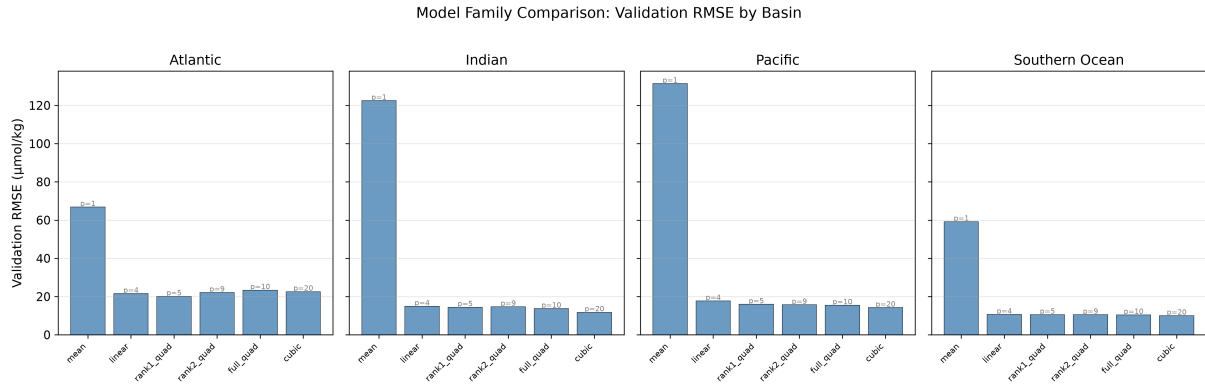


Figure 7: Temporal-validation model hierarchy. Parameter counts are annotated; the rank-1 model has five fitted parameters.

Goyet, C. and Davis, D. (1991). Estimation of total CO₂ concentration throughout the water column. *Deep Sea Research Part A*, 38:871–886.

Key, R. M. et al. (2004). A global ocean carbon climatology: Results from Global Data Analysis Project (GLODAP). *Global Biogeochemical Cycles*, 18.

Lauvset, S. K. et al. (2023). GLODAP v2.2023: the second update of the global ocean data analysis product. *Earth System Science Data*, 15:3661–3685.

Millero, F. J. (1998). Partitioning of CO₂ in the ocean. *Marine Chemistry*, 62:31–47.

Sarmiento, J. L. and Gruber, N. (2006). Ocean biogeochemical dynamics. *Princeton University Press*.

Takahashi, T. et al. (1993). Global air-sea flux of CO₂: An estimate based on measurements of sea-air pCO₂ difference. *Proceedings of the National Academy of Sciences*, 90:11655–11659.